

Coding & Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sivanijana/OptiCrop
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Bootstrap
Key Features Implemented	• Crop Recommendation based on soil and weather data • Machine Learning model for prediction • User-friendly web interface. • Displays suitable crop recommendations.
Pending / Incomplete Features	• Government Scheme Finder • Live weather API integration. • User login and history.
Setup / Run Instructions	1. Install Python and required libraries. 2. Run <code>pip install -r requirements.txt</code>. 3. Run <code>python app.py</code>. 4. Open <code>http://127.0.0.1:5000</code> in a browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop project is developed using Python, Flask, and Machine Learning. It predicts the most suitable crop based on soil nutrients and environmental conditions. The code is organized into separate modules, making it easy to understand, maintain, and improve in the future.